

MODULE 1: Introduction to Database Management and SQL

Course: SWE 306 | Level 300 | Instructor: Miss Precious Oben

1. What is a Database?

A database is an organized collection of data. Think of it like a very organized filing system on a computer. Instead of keeping records in paper files, a database stores them digitally so they can be found quickly and used by many people at the same time.

Every day, people and organizations create large amounts of data. Banks record transactions, hospitals store patient information, and schools keep student records. All of this data needs to be stored somewhere organized so it can be found and used easily. That is what databases are for.

1.1 The Two Main Types of Databases

Relational Databases (SQL): Store data in tables made up of rows and columns. Each row is one record. Each column is one field or attribute. These databases use SQL to manage data.

Non-Relational Databases (NoSQL): Store data in other formats such as documents (like JSON), key-value pairs, or graphs. These are used when data is complex or changes shape often.

Examples of Relational Databases: MySQL, PostgreSQL, Microsoft SQL Server, Oracle Database.

Examples of NoSQL Databases: MongoDB, Redis, Cassandra.

Feature	Relational (SQL)	Non-Relational (NoSQL)
Data format	Tables (rows and columns)	Documents, key-value, graphs
Schema	Fixed and predefined	Flexible
Scalability	Limited	High
Query language	SQL	Own language or API
Best for	Banking, CRM, inventory	Social media, gaming, search

2. Database Management Systems (DBMS)

A Database Management System (DBMS) is software that manages a database. It is the tool that sits between the user and the actual data. It handles storing, retrieving, updating, and deleting data.

There are two kinds of DBMS: Relational (RDBMS) and Non-Relational. In an RDBMS, data is stored in the form of tables.

2.1 Key Components of a DBMS

- **Users:** The people who interact with the database. This includes the database administrator (DBA), developers, and regular end users like customers.
- **Application:** The software that uses the database. This could be a banking app, a hospital system, or a student portal.
- **Storage Engine:** The part that physically saves and reads data from the computer's storage (disk).
- **Metadata Catalog:** Also called a data dictionary. It stores information about the database itself, such as the names of tables, columns, and data types.

- **Database Access Language:** The language used to talk to the database. For relational databases, this is SQL.
- **Optimization Engine:** Takes your query and finds the fastest way to get the result.
- **Query Processor:** Runs the query and returns the results to the user.

3. How Databases Have Evolved

Databases have not always existed in their current form. Before computers, records were kept on paper or cards. As technology developed, better ways to store data were invented.

1. File-Based Systems (before 1960s): Data was stored in flat files. There was no standard way to share data between programs. Problems included duplicate data and no easy way to search.

2. Hierarchical and Network Databases (1960s-1970s): IBM introduced hierarchical databases, which organized data in a tree shape (parent-child relationships). Network databases allowed more complex relationships between data.

3. Relational Databases (1970s-present): E.F. Codd introduced the relational model in 1970. Data is stored in tables and SQL is used to query it. This became the most widely used model. Examples: Oracle, MySQL, SQL Server.

4. NoSQL Databases (2000s-present): The rise of the internet created the need for databases that could handle very large and flexible data. NoSQL databases do not require fixed schemas and can scale very easily.

5. Cloud-Based Databases and DBaaS (2010s-present): Databases moved to the cloud. Services like Amazon RDS and Google Cloud SQL allow organizations to use databases without managing the hardware themselves.

4. Database Architecture

Database architecture describes how the different parts of a database system are arranged and how they communicate. The design affects the speed, security, and reliability of the database.

Single-Tier: Everything (the database, the DBMS, and the user application) runs on one machine. Simple, but not suitable for many users.

Two-Tier: The user application (client) is on one machine and the database is on another (server). The client talks directly to the database.

Three-Tier: A middle layer (application server) sits between the client and the database. This middle layer handles business logic and makes the system more secure and scalable. This is the most common design for web applications.

N-Tier: An extension of three-tier with extra layers such as caching or load balancing. Used for very large applications.

Distributed Architecture: The database is spread across multiple machines or locations. Designed for high availability and global access.

***Note:** Modern databases often run in the cloud and use distributed architecture to serve millions of users at once. Examples include Amazon RDS, Google Cloud SQL, and Azure SQL Database.*

5. Introduction to SQL

SQL (Structured Query Language, pronounced "sequel") is the standard language used to work with relational databases. SQL lets you create databases and tables, insert data, search for data, update records, and delete records.

5.1 Why SQL is Important

- It is fast and efficient for querying large amounts of data.
- It works across many different database systems (MySQL, PostgreSQL, SQL Server, Oracle).
- It is a standard language, which means learning it once lets you work on many platforms.

5.2 SQL Command Categories

DDL (Data Definition Language): Used to create and define database structures. Commands: CREATE, ALTER, DROP, TRUNCATE.

DML (Data Manipulation Language): Used to add, read, update, and delete data. Commands: INSERT, SELECT, UPDATE, DELETE.

TCL (Transaction Control Language): Used to manage transactions (a group of operations that must all succeed or all fail). Commands: COMMIT, ROLLBACK, SAVEPOINT.

DCL (Data Control Language): Used to control access and permissions. Commands: GRANT, REVOKE.

5.3 Basic SQL Syntax

A query is a question you ask the database. The most common query starts with SELECT:

```
SELECT column1, column2 FROM table_name;
```

To get all columns from a table:

```
SELECT * FROM table_name;
```

To filter results:

```
SELECT * FROM table_name WHERE condition;
```

5.4 Tables in SQL

A table is the basic structure for storing data in a relational database. Every table has columns (which define the fields) and rows (which store the actual records).

- Every column must have a data type. Common types: INT (whole numbers), VARCHAR (text), DATE (dates), DECIMAL (numbers with decimal points).
- Columns can have constraints. NOT NULL means the column must always have a value. PRIMARY KEY uniquely identifies each row. UNIQUE means no two rows can have the same value in that column.

Example of creating a table:

```
CREATE TABLE students ( student_id INT PRIMARY KEY, first_name  
VARCHAR(50) NOT NULL, last_name VARCHAR(50) NOT NULL, age INT,  
email VARCHAR(100) UNIQUE );
```

5.5 Database Naming Conventions

When naming databases and tables, follow these guidelines so that your code is clean and easy for others to understand:

- Use clear and descriptive names (e.g., students, not s or data1).
- Avoid SQL reserved words as names (e.g., do not name a table SELECT or ORDER).
- Keep names short but meaningful.
- Be consistent. If you use lowercase, use it everywhere.

- Avoid spaces. Use underscores instead (student_records, not student records).

Summary

In this module, we covered the foundation of database management. A database is an organized collection of data managed by a DBMS. There are two main types: relational (SQL) and non-relational (NoSQL). Databases have evolved from simple flat files in the 1960s to cloud-hosted distributed systems today. SQL is the standard language for relational databases, and it works through four main command categories: DDL, DML, TCL, and DCL. Tables are the core structure, and every column needs a data type and can have constraints.

Next Module: Database Design and Conceptual Modeling (ER Diagrams)